10.5.1

Kavin B-EE25BTECH11033

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Question

Let ABC be a right triangle in which $AB = 6cm, BC = 8cm$ and $\angle B = 90$. BD is the perpendicular from **B** on AC. The circle through $\mathbf{B}, \mathbf{C}, \mathbf{D}$ is drawn. Construct the tangents from **A** to this circle.

Theoretical Solution

Let,

$$\mathbf{B} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \mathbf{A} = \begin{pmatrix} 0 \\ 6 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 8 \\ 0 \end{pmatrix} \tag{1}$$

Let the equation of the line passing through **A** and **C** be,

$$\mathbf{n}^\top \mathbf{x} = c \tag{2}$$

The direction vector of line AC is given by,

$$\mathbf{m} = \mathbf{A} - \mathbf{C} \equiv \begin{pmatrix} -4 \\ 3 \end{pmatrix} \tag{3}$$

Therefore, the normal vector of the desired line is given by,

$$\mathbf{n} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \tag{4}$$

$$c = \mathbf{n}^\top \mathbf{A} = \begin{pmatrix} 3 & 4 \end{pmatrix} \begin{pmatrix} 0 \\ 6 \end{pmatrix} = 24 \tag{5}$$

Theoretical Solution

Let $\mathbf{D}$ be the foot of the perpendicular from $\mathbf{B}$ to the line

$$\begin{pmatrix}\mathbf{m} & \mathbf{n}\end{pmatrix}^\top \mathbf{D} = \begin{pmatrix}\mathbf{m}^\top\mathbf{B} \\ c\end{pmatrix} \tag{6}$$

$$\begin{pmatrix}-4 & 3 \\ 3 & 4\end{pmatrix} \mathbf{D} = \begin{pmatrix}\mathbf{m}^\top\mathbf{B} \\ 24\end{pmatrix} \tag{7}$$

$$\mathbf{m}^\top\mathbf{B} = \begin{pmatrix}-4 & 3\end{pmatrix}\begin{pmatrix}0 \\ 0\end{pmatrix} = 0 \tag{8}$$

$$\implies \begin{pmatrix}-4 & 3 \\ 3 & 4\end{pmatrix} \mathbf{D} = \begin{pmatrix}0 \\ 24\end{pmatrix} \tag{9}$$

Theoretical Solution

The augmented matrix is given by,

$$\left(\begin{array}{cc|c} -4 & 3 & 0 \\ 3 & 3 & 24 \end{array}\right) \tag{10}$$

$$R_1 \to \frac{-1}{4} R_1 \implies \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 3 & 3 & 24 \end{array}\right) \tag{11}$$

$$R_2 \to R_2 - 3R_1 \implies \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 25/4 & 24 \end{array}\right) \tag{12}$$

$$R_2 \to \frac{4}{25} R_2 \implies \left(\begin{array}{cc|c} 1 & -3/4 & 0 \\ 0 & 1 & 96/25 \end{array}\right) \tag{13}$$

$$R_1 \to R_1 + \frac{3}{4} R_2 \implies \left(\begin{array}{cc|c} 1 & 0 & 72/25 \\ 0 & 1 & 96/25 \end{array}\right) \tag{14}$$

$$\begin{pmatrix} 72/25 \end{pmatrix}$$

Theoretical Solution

To find the circle through $\mathbf{B}, \mathbf{C}, \mathbf{D}$ the parameters are given by

$$\begin{pmatrix} 2\mathbf{B} & 2\mathbf{C} & 2\mathbf{D} \\ 1 & 1 & 1 \end{pmatrix}^{\top} \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = -\begin{pmatrix} \|\mathbf{B}\|^2 \\ \|\mathbf{C}\|^2 \\ \|\mathbf{D}\|^2 \end{pmatrix} \tag{16}$$

$$\begin{pmatrix} 0 & 0 & 1 \\ 16 & 0 & 1 \\ 144/25 & 192/25 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{u} \\ f \end{pmatrix} = \begin{pmatrix} 0 \\ -64 \\ -576/25 \end{pmatrix} \tag{17}$$

The augmented matrix is given by,

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 0 \\ 16 & 0 & 1 & -64 \\ 144/25 & 192/25 & 1 & -576/25 \end{array}\right) \tag{18}$$

$$R_1 \leftrightarrow R_2 \implies \left(\begin{array}{ccc|c} 16 & 0 & 1 & -64 \\ 0 & 0 & 1 & 0 \\ 144/25 & 192/25 & 1 & -576/25 \end{array}\right) \tag{19}$$

Theoretical Solution

$$R_3 \to R_3 - \frac{144}{25}R_1 \implies \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 0 & 1 & 0 \\ 0 & 192/25 & 16/25 & -504/25 \end{array}\right) \tag{21}$$

$$R_2 \leftrightarrow R_3 \implies \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 192/25 & 16/25 & -504/25 \\ 0 & 0 & 1 & 0 \end{array}\right) \tag{22}$$

$$R_2 \to \frac{25}{192}R_2 \implies \left(\begin{array}{ccc|c} 1 & 0 & 1/16 & -4 \\ 0 & 1 & 1/12 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \tag{23}$$

$$R_2 \to R_2 - \frac{1}{12}R_3\ ,\ R_1 \to R_1 - \frac{1}{16}R_3 \implies \left(\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \tag{24}$$

Theoretical Solution

Let $\mathbf{n}^\top \mathbf{x} = c$ be the equation of tangent to the circle.
Also,

$$\mathbf{n}^\top \mathbf{A} = c \tag{26}$$

and,

$$\frac{\left|\mathbf{n}^\top(-\mathbf{u}) - c\right|}{\|\mathbf{n}\|} = r \tag{27}$$

where

$$r = \sqrt{\|\mathbf{u}\|^2 - f}$$

$$\implies \mathbf{n}^\top \mathbf{u} + c = \pm \|\mathbf{n}\| \sqrt{\|\mathbf{u}\|^2 - f} \tag{28}$$

$$\implies \mathbf{n}^\top = (\mathbf{u} + \mathbf{A}) \pm \|\mathbf{n}\| \sqrt{\|\mathbf{u}\|^2 - f} \tag{29}$$

Theoretical Solution

substitute $\mathbf{n} = \begin{pmatrix} -m \\ 1 \end{pmatrix}$ where m is the slope of the line.

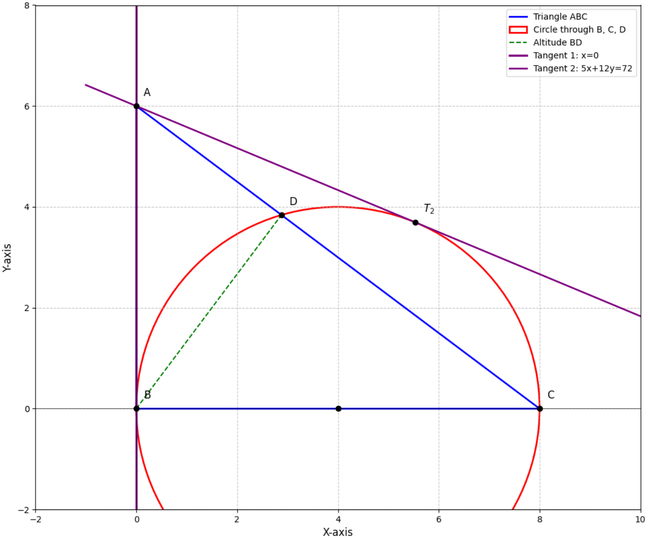
$$\implies \begin{pmatrix} -m & 1 \end{pmatrix} \begin{pmatrix} -4 \\ 6 \end{pmatrix} = \pm 4\sqrt{1 + m^2} \tag{30}$$

$$m = -5/12 \implies \mathbf{n} = \begin{pmatrix} 5 \\ 12 \end{pmatrix} \tag{31}$$

$$c = \mathbf{n}^\top \mathbf{A} = 72 \tag{32}$$

Therefore, the line $\begin{pmatrix} 5 & 12 \end{pmatrix} \mathbf{x} = 72$ and the $\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 0$ are the two tangents.

Plot



Triangle ABC
Circle through B, C, D
Altitude BD
Tangent 1: x=0
Tangent 2: 5x+12y=72
A
D
T_2
B
C
X-axis
Y-axis
−2
0
2
4
6
8
10